

Fluid Flow in Pipes

CHE 381 - Chemical Engineering Lab I
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Introduction



Purpose of the lab:

- Finding how the pressure drop changes in the orifice, elbow, bend, expansions and contractions. (also all the small, medium, and large pipes)
- How pressure drop is affected by diameter, flow rate, and different pipe material
- Using these values we can calculate the head loss, friction factor and flow coefficient.

Theory



Bernoulli's equation is used for laminar flow to calculate the pressure drop:

$$\Delta p = p_1 - p_2 = \frac{1}{2}\rho V_2^2 - \frac{1}{2}\rho V_1^2$$



Pressure loss for flows that are laminar and turbulent:

$$h_L = \frac{\Delta P}{\rho} = f \frac{L V^2}{D 2}$$

For turbulent flows, the friction factor is given by the Colebrook Equation, which is defined as follows:

$$\frac{1}{\sqrt{f}} + 1.737 \ln \left\{ \frac{1.256}{\text{Re} \sqrt{f}} + \frac{k}{3.707 D} \right\} = 0$$



Theory

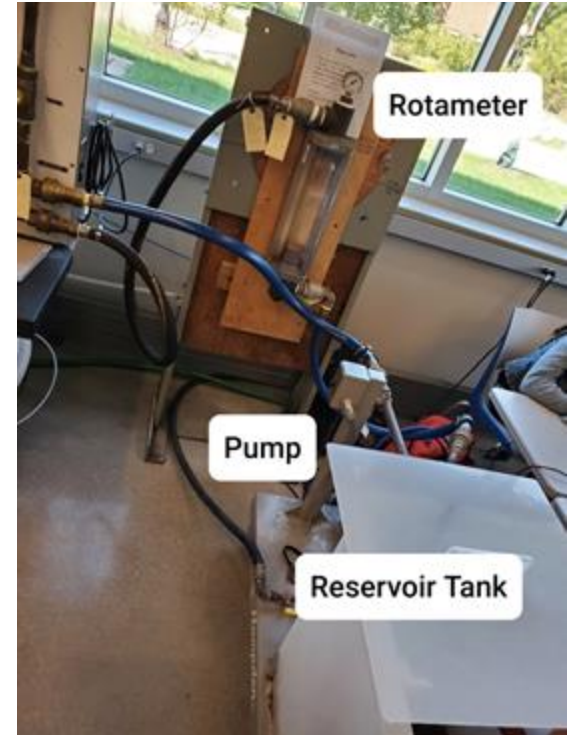
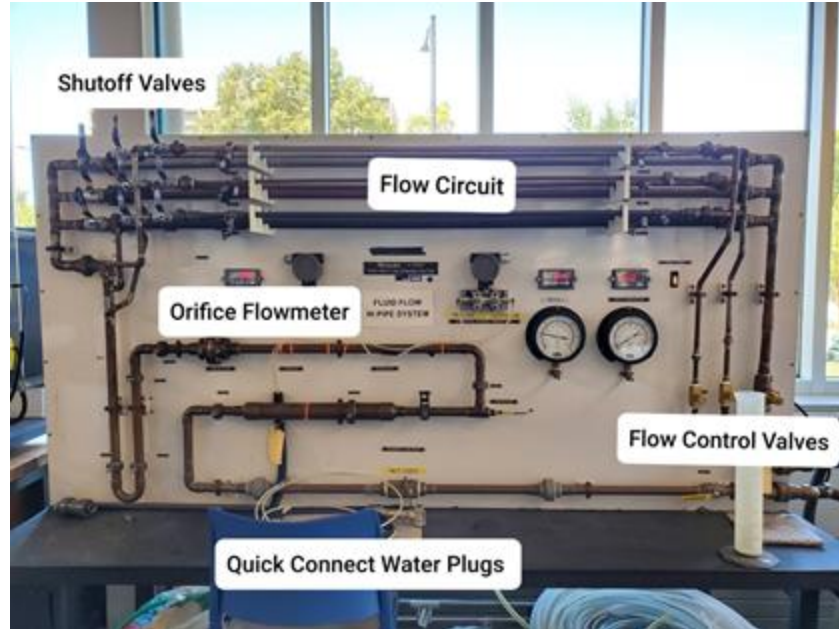
The loss coefficient is given by the following equation:

$$h_L = K \frac{v^2}{2}$$

Where the loss coefficient, K, is equal to $K = f \frac{L}{D}$ (Equation 7).



Apparatus





Material & Supplies



- Water
- 1000 mL Graduated Cylinder
- Quick Connect Plugs
- Stopwatch



Procedure



- Fill the reservoir tank with water past the pumps bypass tube.
- Activate the pump and open the flow valve to let water flow into the system.
- Calibrate the rotameter by measuring time taken to fill 1000-mL graduated cylinder and comparing to rotameter reading.
- Measure pressure drop across fittings, orifice plate, bends, and different piping using quick-connect plugs.



Experimental Challenges



- Inconsistent pressure difference from gauge after using same connect plugs
- Inconsistent flow rate due to eye measurement of rotameter



Safety



Electrical Hazards

- possibility for electricity and water to come in contact

Mechanical Hazards

- possibility of valve failure and high pressure water to face/eyes

Pressure Hazards

- pumps exceeding recommended operating conditions have potential to fail



References

-Lab Hanout for Fluid Flow in Pipes

Comment Summary

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1. Date?

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2. 97/100

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4. How was this modified for turbulent flows?

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6. Don't paste equations

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